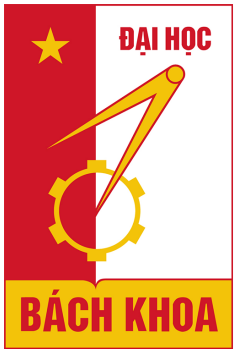
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY





Report:

Project II: Comparing Training-Free Search Strategies for Multimodal Document Question-Answering with ColQwen2

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Mục lục

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Abstract

Multimodal document question answering (DocVQA) requires retrieving the specific document image that contains the answer to a natural language query, followed by extracting or generating that answer from visual and textual cues within the document. Recent dense retrieval models such as ColQwen2, which employ late-interaction MaxSim scoring over patch-level embeddings, achieve strong retrieval accuracy without requiring OCR pre-processing, but remain imperfect when queries reference fine-grained textual details that are better captured by lexical matching. This work presents a systematic, training-free comparison of retrieval strategies for multimodal DocVQA, evaluated entirely through inference-time composition of existing pretrained models rather than fine-tuning. We benchmark four retrieval configurations: **dense retrieval** using ColQwen2 alone, **sparse retrieval** using BM25 over text extracted via docTR OCR, **Reciprocal Rank Fusion (RRF)** combining the dense and sparse ranked lists, and **VLM-based re-ranking**, in which a vision-language model (Qwen2-VL) directly scores document-query relevance via prompted yes/no log-probabilities over the dense retriever's top candidates. Experiments are conducted on 2,000 validation queries from the DocVQA dataset against a deduplicated corpus of 10,189 unique document images. Retrieval quality is measured using Recall@1/5/10, MRR, and NDCG@10. We further analyze the latency–accuracy trade-off of VLM re-ranking relative to its added inference cost. Our results provide practical guidance on selecting retrieval strategies for multimodal RAG systems under different accuracy and latency constraints, without requiring any additional model training.

1 Introduction

1.1 Motivation

Document Visual Question Answering (DocVQA) involves answering natural language questions about the content of scanned or photographed documents – forms, invoices, letters, and reports – where the answer may be embedded in text, tables, or layout structure. In a retrieval-augmented generation (RAG) setting, this task decomposes into two stages: (1) **retrieval**, identifying the document image most likely to contain the answer from a large corpus, and (2) **generation**, extracting or synthesizing the answer conditioned on the retrieved document and query.

The retrieval stage is the primary bottleneck for end-to-end system accuracy: if the wrong document is retrieved, even a perfect generation model cannot recover the correct answer. Recent multi-vector, late-interaction retrieval models – most notably ColPali and its successor ColQwen2 – have substantially advanced document retrieval by embedding image patches directly, bypassing the need for OCR while capturing fine-grained visual-textual alignment via MaxSim scoring. However, purely dense retrieval can still underperform on queries that hinge on exact textual matches (e.g., specific numbers, dates, or named entities), a known weakness of dense embeddings relative to lexical methods such as BM25.

1.2 Problem Statement

Given the complementary strengths of dense (visual-semantic) and sparse (lexical) retrieval, a natural question arises: **can retrieval performance for multimodal DocVQA be improved by combining dense and sparse signals, or by applying a stronger semantic judge at re-ranking time, without any additional model training?** This project investigates that question by comparing three training-free combination strategies built on top of a ColQwen2 dense baseline, all operating purely at inference time:

- **Sparse retrieval as a standalone alternative**, using BM25 over OCR text extracted by docTR, to test how well a purely lexical signal performs on its own;
- **Score/rank fusion** via Reciprocal Rank Fusion (RRF), combining the independently computed dense (ColQwen2) and sparse (BM25) rankings;
- **Post-retrieval re-ranking with a vision-language model**, where ColQwen2 produces an initial candidate shortlist that is then re-scored by Qwen2-VL's semantic judgment of document-query relevance.

Each strategy represents a different trade-off between retrieval accuracy, computational cost, and implementation complexity, and no existing study – to our knowledge – systematically compares all of these strategies under a common experimental protocol specifically using ColQwen2 as the base dense retriever.

1.3 Project Objectives

To answer this question, this project aims to design, implement, and empirically evaluate a training-free multimodal retrieval pipeline for DocVQA. The specific objectives include:

- **Dense and Sparse Retrieval Implementation:** To build a ColQwen2-based dense retrieval pipeline using late-interaction MaxSim scoring, and a complementary sparse retrieval pipeline using docTR for OCR text extraction and BM25 for lexical indexing, both evaluated independently as baselines.
- **Hybrid Retrieval via Rank Fusion:** To combine the dense and sparse ranked lists using Reciprocal Rank Fusion (RRF), and quantify whether this fusion yields consistent improvements in Recall@K, MRR, and NDCG@10 over either retriever alone.
- **Semantic Re-ranking with a Vision-Language Model:** To apply Qwen2-VL as a post-retrieval re-ranker over ColQwen2's top candidates, using prompted yes/no log-probability scoring, and to characterize the resulting accuracy gains against the additional inference latency incurred.

2 Related Work

The development of a training-free multimodal retrieval pipeline for DocVQA intersects with four separate lines of research: multi-vector dense retrieval for visually rich documents, sparse lexical retrieval, rank/score fusion, and the use of vision-language models as relevance judges. This section reviews each in turn and situates our work relative to them.

2.1 Multi-vector Dense Retrieval for Documents

Conventional document retrieval pipelines index a page's textual content after an OCR pass, discarding the spatial and visual structure that carries meaning in tables, forms, and figures, since OCR-based PDF retrieval breaks on real documents because OCR discards the spatial relationships between tables, charts, and text. **ColPali** was proposed to address this limitation by treating each document page directly as an image and producing high-quality multi-vector patch embeddings from it, combined with a late interaction matching mechanism that largely outperforms modern document retrieval pipelines relying on OCR-extracted text. Architecturally, ColPali encodes a page with a vision-language backbone (originally PaliGemma) and projects each resulting patch embedding to a shared 128-dimensional space, preserving patch-level granularity rather than pooling visual features into a single global vector.

Relevance is computed using the **MaxSim** operator inherited from ColBERT's late-interaction paradigm, which sums, for every query token, the maximum similarity against all document patch embeddings, enabling fine-grained matching while remaining computationally tractable at retrieval scale. **ColQwen2** extends this framework by replacing the PaliGemma backbone with a Qwen2-based vision-language model, giving rise to the name Contextualized Late Interaction over Qwen2. Independent reproduction studies confirm that this backbone substitution is not incidental: upgrading the underlying VLM backbone from PaliGemma to Qwen2-VL yields a consistent effectiveness gain, with the reproduced ColQwen2 outperforming the reproduced ColPali on nDCG@5. This finding directly motivates our choice of ColQwen2, rather than the original ColPali, as the dense retrieval backbone for this project.

Despite these advances, multi-vector dense retrieval is not uniformly reliable across query types. Analysis of the query-token-matching behavior of ColPali-family models reveals that

reducing lexical matching impacts retrieval effectiveness substantially more than reducing non-lexical matching, indicating that these models rely heavily on lexical token overlap when it is available and only fall back to non-lexical (purely visual-semantic) matching when it is not. This observation is central to the motivation of our work: because dense retrieval implicitly depends on lexical cues to achieve its best performance, an explicit, OCR-driven lexical signal (Section 2.2) may still offer complementary value even to a strong late-interaction retriever such as ColQwen2.

2.2 Sparse Lexical Retrieval and OCR-based Indexing

Prior to the emergence of visual document embedding models, the dominant approach to document retrieval was to extract text via Optical Character Recognition (OCR) and index it using classical sparse (lexical) retrieval methods such as **BM25**. BM25 scores a document by aggregating term-frequency and inverse-document-frequency statistics over the query's terms, and remains a strong, training-free baseline for queries that hinge on exact lexical matches – numbers, dates, proper nouns, and other named entities – which dense embedding models can struggle to represent precisely. Because DocVQA questions frequently reference such fine-grained textual details (e.g., a specific total on an invoice or a date on a form), a sparse retriever built on OCR-extracted text remains a relevant and complementary baseline even in the era of end-to-end visual document embeddings. In this project, OCR text is extracted using **docTR**, an open-source deep-learning-based OCR toolkit, and indexed with the standard `rank_bm25` implementation of BM25 to construct the sparse retrieval component of our pipeline.

2.3 Rank and Score Fusion Methods

Given two independently produced ranked lists – one dense, one sparse – a natural training-free strategy for combining their complementary strengths is **rank fusion**. **Reciprocal Rank Fusion (RRF)** is a widely adopted, parameter-light fusion method that re-scores each document according to the reciprocal of its rank in each input list, summed across lists, rather than attempting to normalize and combine raw similarity scores that may lie on incomparable scales (e.g., a BM25 score versus a MaxSim cosine-based score). This rank-based formulation makes RRF particularly attractive in our setting, since ColQwen2's MaxSim scores and BM25's term-weighted scores are not directly comparable in magnitude or distribution, and RRF sidesteps the need for any score normalization or learned fusion weights – preserving the training-free constraint of this project while still allowing the sparse retriever's lexical precision to correct cases where the dense retriever's ranking is imperfect.

2.4 Vision-Language Models as Re-rankers

A complementary, training-free strategy for improving retrieval precision is **post-retrieval re-ranking**, in which a cheap first-stage retriever produces a candidate shortlist that is subsequently re-scored by a more expressive, but more computationally expensive, second-stage model. Large vision-language models (VLMs) such as **Qwen2-VL** are increasingly used in this capacity: rather than being trained explicitly as retrievers, they are prompted to directly judge document-query relevance (e.g., answering whether a given document image plausibly answers a given question)

and the model's confidence in an affirmative response, read off as the log-probability of a specific token, is used as a relevance score. This approach leverages the VLM's broad multimodal reasoning capability to correct cases where a purely embedding-based similarity metric (dense or sparse) misranks a document, at the cost of substantially higher per-candidate inference latency – making it best suited to re-ranking a small shortlist (e.g., the top-20 candidates from a first-stage retriever) rather than scoring an entire corpus. This latency-accuracy trade-off is a central axis of evaluation in our experiments (Section 5.2).

2.5 Positioning of This Work

While dense retrieval with ColPali-family models, sparse OCR-based retrieval, rank fusion, and VLM-based re-ranking have each been studied independently, no existing work – to our knowledge – systematically compares all four strategies under a single, common, training-free protocol specifically built on ColQwen2 as the dense retrieval backbone, while also comprehensively evaluating retrieval performance across standard metrics. This project addresses that gap.

3 Dataset

3.1 DocVQA Dataset Overview

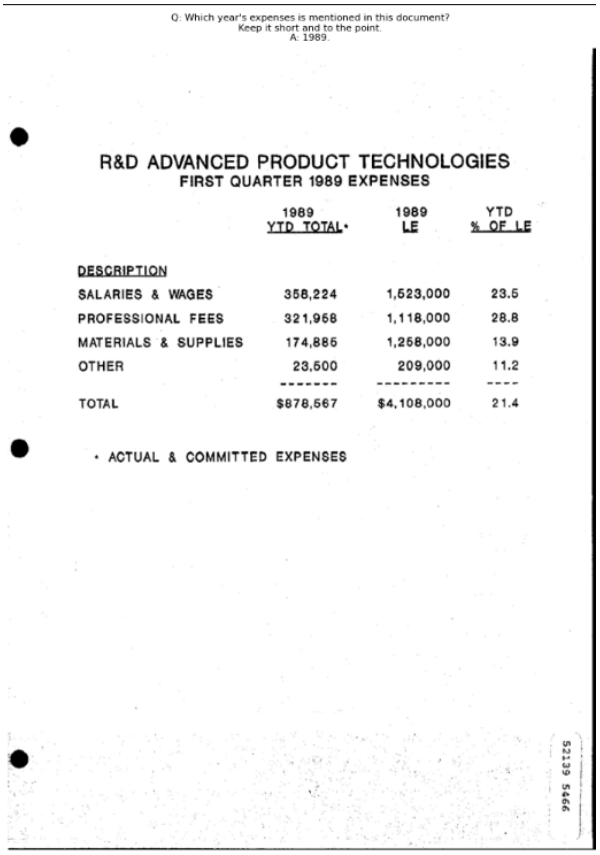
This project uses the **DocVQA** dataset, a benchmark for visual question answering on document images. The dataset comprises **37,463 training** and **2,000 validation** question-answer pairs drawn from **6,071 industry documents** sourced from the UCSF Industry Documents Library, spanning five major industries: tobacco, food, drug, fossil fuel, and chemical. Document images are hand-picked to prioritize visually rich layouts such as tables, forms, and lists, with content ranging from 1900 to 2018.

Document Characteristics: DocVQA pages contain a diverse mix of printed, typewritten, and handwritten content, including letters, memos, notes, and reports. Unlike text-only datasets, each document image is visually complex, combining text regions, numerical data, tables, and spatial layout patterns that carry semantic meaning. This visual-structural diversity makes DocVQA well-suited for evaluating retrieval approaches that leverage both visual and textual signals.

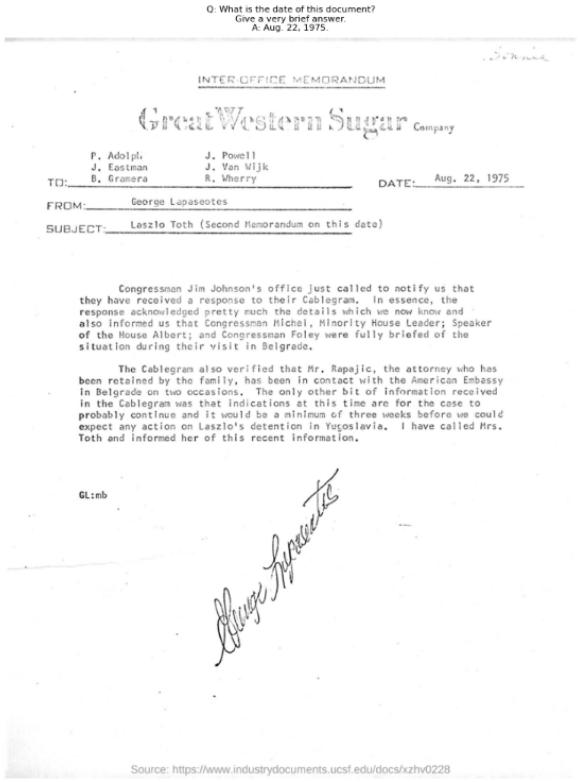
Annotation and Answer Format: Questions were collected through web-based annotation where workers were shown a document image and asked to define question-answer pairs, with answers entered verbatim from the document. This makes DocVQA fundamentally an *extractive* task. As a result, ground-truth answers are predominantly numeric values (totals, dates, quantities) and specific text spans (names, codes), reflecting the prevalence of invoices, forms, and financial reports in the source corpus. This characteristic directly motivates inclusion of a lexical (BM25) retrieval signal alongside dense methods, since exact numeric and textual matches are precisely where sparse retrieval excels.

3.2 Example Documents

Figure 1 shows two representative DocVQA documents illustrating the diversity of document types and layout complexity in the dataset.



(a) Financial report with structured tables



(b) Memorandum with handwritten annotation

Hình 1: Representative DocVQA documents: (a) R&D expense report with tabular financial data and numeric values; (b) business memorandum combining printed and handwritten annotations. Both exemplify the visual and textual complexity characteristic of DocVQA documents.

3.3 Data Structure

Each validation record in DocVQA consists of the following fields:

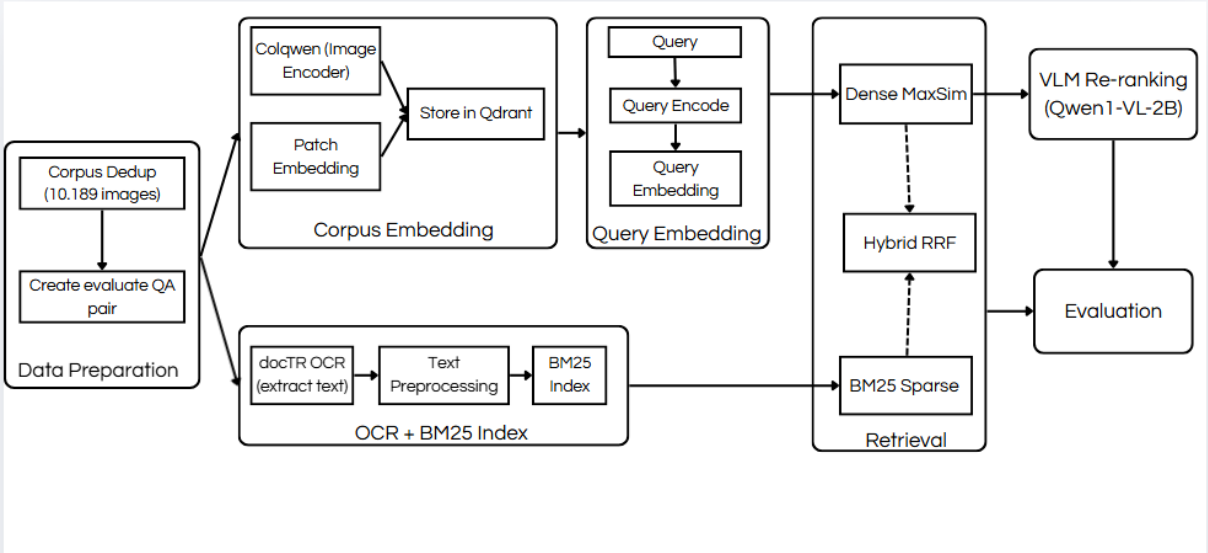
- **Query:** A natural language question (e.g., "Which year's expenses are mentioned in this document?").
- **Image filename:** A unique identifier for the document image containing the answer.
- **Ground-truth answer:** The correct answer as extracted verbatim from the document.
- **Document image:** A scanned or photographed page of the source document at fixed resolution.

For retrieval experiments, only the validation split (2,000 queries) is used, consistent with the training-free scope of this project (Section 4). The training split is not leveraged for any model fine-tuning or retrieval index construction.

4 Methodology

4.1 System Pipeline Overview

Figure 2 presents the overall architecture of the proposed training-free multimodal document retrieval pipeline. The system consists of five main stages, beginning with corpus preparation and ending with retrieval evaluation. The entire pipeline operates using pretrained models only, without any additional fine-tuning.



Hình 2: Overview of the proposed training-free multimodal retrieval pipeline. The system combines dense retrieval using ColQwen2, sparse retrieval using OCR+BM25, hybrid retrieval via Reciprocal Rank Fusion (RRF), and VLM-based re-ranking for multimodal document retrieval.

The pipeline begins with the **Data Preparation** stage, where duplicate document images are removed from the DocVQA corpus to construct a retrieval corpus containing 10,189 unique document images. At the same time, the corresponding question–answer pairs are retained for later evaluation.

Next, the corpus is processed through two parallel indexing branches. In the **Corpus Embedding** branch, each document image is encoded by ColQwen2 into patch-level embeddings, which are stored in a Qdrant vector database for efficient dense retrieval. In parallel, the **OCR + BM25 Index** branch extracts textual content from each document using docTR OCR, performs text preprocessing, and builds a BM25 lexical index for sparse retrieval.

During inference, an input query is encoded by the ColQwen2 query encoder to produce query embeddings. These embeddings are matched against the document embeddings using the Dense MaxSim operator to obtain dense retrieval scores. Simultaneously, the textual query is searched against the BM25 index to produce sparse retrieval results.

The dense and sparse rankings can be used independently or combined through **Reciprocal Rank Fusion (RRF)** to exploit the complementary strengths of semantic and lexical retrieval. Finally, the highest-ranked candidates are optionally passed to a **Qwen2-VL-2B** vision-language model, which performs semantic re-ranking by directly judging the relevance between the query

and candidate document images.

The final ranked results are evaluated using standard retrieval metrics such as Recall@K, Mean Reciprocal Rank (MRR), and NDCG@10. This modular design enables a systematic comparison of dense retrieval, sparse retrieval, hybrid retrieval, and VLM-based re-ranking under a unified, training-free experimental framework.

4.2 Data Preparation and Corpus Construction

4.2.1 Corpus Deduplication

DocVQA is structured as a question-answer dataset where multiple questions may reference the same document image. To construct a well-defined retrieval corpus, all document images in the validation split are deduplicated by their `image_filename` field, retaining exactly one copy of each unique image. This deduplication step reduces the validation split from 2,000 question-image pairs to a corpus of **10,189 unique document images**, which serves as the fixed retrieval index for all experiments.

4.2.2 Evaluation Query Set and Ground Truth

For each of the 2,000 validation questions, the associated ground-truth document image identifier and answer text are retained, forming a set of 2,000 `(query, image_id, answer)` triples. Retrieval success at rank k is measured as whether the ground-truth `image_id` appears within the top-k returned documents, forming the basis of Recall@k, MRR, and NDCG@10 metrics (Section 5.1).

4.2.3 Input Resolution and Preprocessing

All document images are resized to a fixed resolution constraint prior to embedding, with `fixed_pixels` $= 512 \times 28 \times 28$, ensuring consistent input dimensions across all stages and avoiding excessive memory consumption on GPU resources.

4.3 Corpus Embedding with ColQwen2

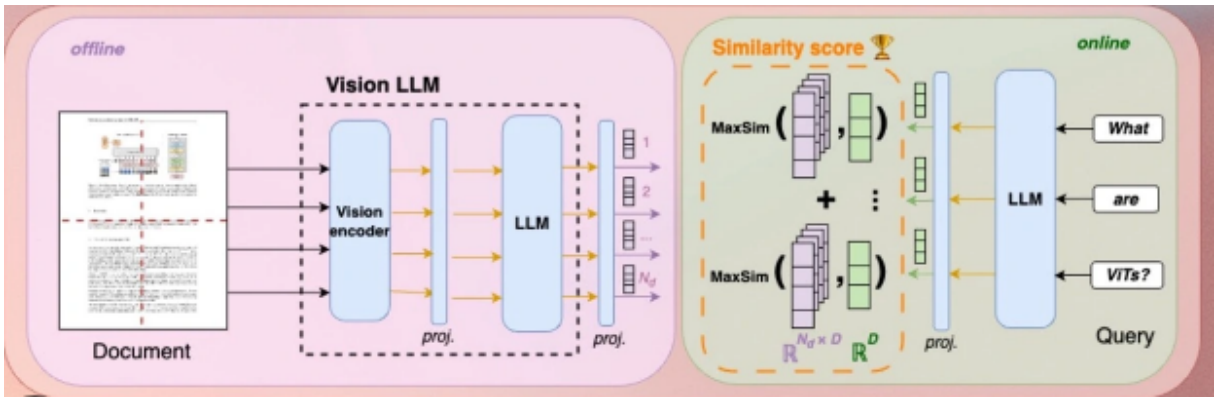
ColQwen2 is a multi-vector, late-interaction dense retrieval model trained on document image-query pairs. For each of the 10,189 unique document images in the corpus, we compute its patch-level embeddings as follows:

1. The document image is passed through the Qwen2-VL vision backbone, which divides the image into a grid of non-overlapping patches and produces a contextual embedding for each patch.

2. Each patch embedding is projected to a shared 128-dimensional space via a learned linear projection layer.

3. The complete set of patch embeddings $\mathbf{D} = \{\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m\}$ (where m is the number of patches in the document) is stored in a dense vector index.

This patch-level representation preserves fine-grained visual and textual information within the document, enabling precise matching at retrieval time without requiring OCR preprocessing.



Hình 3: Overview of the ColPali/ColQwen2 late-interaction retrieval framework. Offline, document images are encoded into multi-vector patch embeddings. During online retrieval, query token embeddings are matched against document patch embeddings using the MaxSim operator to compute relevance scores.

Figure 3 illustrates the retrieval framework adopted by ColPali and inherited by ColQwen2. During the offline stage, each document image is encoded by a vision-language model into multiple patch-level embeddings that are stored in the retrieval index. During online retrieval, the input query is encoded into token-level embeddings, and the MaxSim operator computes the relevance score by matching each query token with its most similar document patch. This late-interaction design preserves fine-grained visual-textual information while remaining efficient for large-scale retrieval.

4.4 Query Embedding with ColQwen2

Unlike corpus embedding, which is performed offline only once for all document images, query embedding is executed online during inference for every incoming user query. The same pretrained ColQwen2 retrieval model is used to ensure that both queries and document images are represented in the same embedding space.

For each natural language query, the ColQwen2 processor first tokenizes the input text and converts it into model inputs with appropriate padding. The query is then passed through the ColQwen2 retrieval model, which produces contextualized token-level embeddings rather than a single global sentence representation. Let the encoded query be represented as

$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n\},$$

where each $\mathbf{q}_i \in \mathbb{R}^{128}$ denotes the embedding of one valid query token after padding tokens have been removed, and n is the number of query tokens.

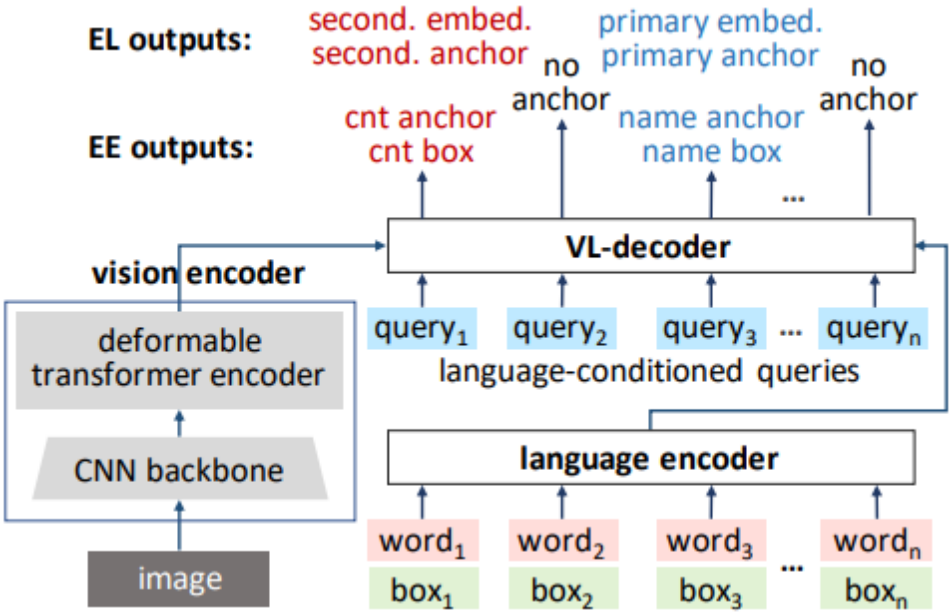
Because ColQwen2 follows the late-interaction retrieval paradigm inherited from ColBERT, every query token embedding is preserved instead of being pooled into a single vector. This enables each query token to independently match the most relevant visual patch in a document during retrieval.

To improve retrieval efficiency, embeddings for the 2,000 evaluation queries are computed once and stored as NumPy object arrays together with their corresponding metadata (query text, ground-truth document identifier, and answer). These cached embeddings are reused throughout all retrieval experiments, avoiding repeated encoding while ensuring identical query representations across different retrieval strategies.

4.5 Sparse Retrieval via OCR and BM25

To construct a sparse lexical retrieval baseline, we extract text from each document image and build a BM25 index as follows:

1. **OCR Extraction:** Each of the 10,189 unique document images is processed by the docTR OCR toolkit to extract plain text. docTR is a deep-learning-based OCR system designed for document analysis, capable of handling both printed and handwritten content while preserving document structure.



Hình 4: Simplified architecture of docTR. The model combines a CNN-based visual encoder with a Transformer decoder conditioned on textual queries to jointly detect text regions and recognize their corresponding content in an end-to-end manner.

Figure 4 illustrates the overall architecture of docTR. The framework consists of a CNN backbone and a deformable Transformer encoder that extract visual features from the input document image. These features are then decoded by a language-aware Transformer decoder to simultaneously predict text locations and recognize the corresponding text content in an end-to-end fashion. This unified architecture enables docTR to efficiently handle both printed and handwritten document images for OCR extraction.

2. **Text Preprocessing:** Extracted text is lowercased, and standard punctuation normalization is applied to ensure consistent matching during retrieval.

3. **BM25 Indexing:** The preprocessed text for each document is indexed using the `rank_bm25` Python implementation of BM25, a probabilistic retrieval model that scores documents by combining term frequency (TF) and inverse document frequency (IDF) statistics.

BM25 is parameter-free and training-free, making it an ideal baseline for lexical matching in this work. At retrieval time, each query is tokenized and matched against the BM25 index to produce a ranked list of documents.

4.6 Retrieval Stage: Four Strategies

For each of the 2,000 queries, we apply four retrieval strategies independently, each producing a ranked list of candidate documents.

4.6.1 Strategy 1: Dense Retrieval (ColQwen2)

Dense retrieval encodes the query using the same Qwen2-VL backbone as the corpus embedding stage, producing query token embeddings

$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n\}.$$

Unlike conventional dense retrieval models that pool token representations into a single vector, ColQwen2 preserves token-level embeddings and employs the late-interaction MaxSim operator inherited from ColBERT. Relevance between a query and a document is computed as:

$$\text{score}(\mathbf{Q}, \mathbf{D}) = \sum_{i=1}^{n} \max_{j=1}^{m} \cos(\mathbf{q}_i, \mathbf{d}_j),$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity, n is the number of valid query tokens, and m is the number of document patch embeddings. For each query token, only the highest similarity among all document patches contributes to the final score, enabling fine-grained matching between textual concepts and localized visual-textual regions within the document.

Since different queries contain different numbers of tokens, ColQwen2 preserves the original query length instead of forcing all queries to a fixed size. Padding tokens do not carry semantic information and may distort similarity calculations; therefore, they are removed using the attention mask before MaxSim computation. As a result, only valid query tokens participate in the retrieval process.

For each query, the MaxSim score is computed independently against every document in the corpus. Documents are ranked in descending order of their MaxSim scores, producing a ranked list $\text{Rank}_{\text{dense}}$, and the highest-scoring document is selected as the Top-1 retrieval result.

Because raw MaxSim scores increase with the number of query tokens, they are not directly comparable across different queries. To obtain a normalized similarity measure, the Top-1 MaxSim score is divided by the number of valid query tokens:

$$\text{Mean-MaxSim} = \frac{\text{MaxSim}(\mathbf{Q}, \mathbf{D})}{n}.$$

Finally, an overall embedding-similarity statistic is computed across the validation set as:

$$\text{Average MaxSim} = \frac{1}{N} \sum_{i=1}^{N} \text{Mean-MaxSim}_i,$$

where N is the total number of evaluation queries. In our experiments, the resulting **Average MaxSim is 0.78844**. This value reflects the average embedding similarity between queries and their highest-ranked retrieved documents. It is reported as a complementary indicator of embedding quality, while Recall@K, MRR, and NDCG remain the primary retrieval evaluation metrics.

4.6.2 Strategy 2: Sparse Retrieval (BM25 over OCR)

Sparse retrieval applies BM25 directly to the query text (after lowercasing and tokenization) against the BM25 index constructed from OCR-extracted document text (Section 4.5). The BM25 score is computed as:

$$\text{BM25}(q, D) = \sum_{i=1}^{n} \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{L_{\text{avg}}}\right)}$$

where $f(q_i, D)$ is the frequency of query term q_i in document D, $|D|$ is the document length, L_{avg} is the average document length in the corpus, and k_1 and b are hyperparameters (typically $k_1 = 1.5, b = 0.75$). Documents are ranked by descending BM25 score, producing a ranked list $\text{Rank}_{\text{sparse}}$.

4.6.3 Strategy 3: Hybrid Retrieval via Reciprocal Rank Fusion (RRF)

Reciprocal Rank Fusion combines the independently computed dense and sparse ranked lists by re-scoring documents based on the reciprocals of their ranks in each list:

$$\text{RRF_score}(d) = \sum_{k \in \{\text{dense}, \text{sparse}\}} \frac{1}{K + \text{rank}_k(d)}$$

where K is a constant (we set $K = 60$), and $\text{rank}_k(d)$ is the rank position of document d in the ranked list from retriever k. If a document does not appear in a ranked list (treated as rank $= \infty$), it contributes zero to the sum. Documents are then ranked by descending RRF score, producing a merged ranked list Rank_{RRF}. This rank-based fusion is training-free and avoids the need to normalize raw scores that lie on incomparable scales.

4.6.4 Strategy 4: Dense Retrieval with VLM Re-ranking

In this strategy, ColQwen2 produces an initial dense-retrieval ranked list, and the top-N candidates (e.g., $N = 20$) are re-scored by a vision-language model (Qwen2-VL) acting as a learned semantic judge.

1. Retrieve top-N documents from the dense retrieval stage using ColQwen2.
2. For each top-N document and the query, prompt Qwen2-VL with the following template:

```
"Does the following document answer the question: [QUERY]?
Document: [IMAGE]
Answer yes or no."
```

3. Extract the log-probability of the token "yes"from the model's output distribution as the relevance score for that document.
4. Re-rank the top-N documents by descending log-probability score, producing a re-ranked list Rank_{VLM}.

This re-ranking leverages the VLM's broad multimodal reasoning to correct misranked documents from the dense retriever, at the cost of substantially higher inference latency (discussed in Section 5.2).

4.7 Experimental Setup

4.7.1 Computational Environment

All experiments are conducted on a single NVIDIA GPU (T4 or P100) with 16 GB memory, using PyTorch 2.0 as the deep learning framework. ColQwen2 embeddings for the corpus (10,189 documents) are computed once and cached in memory or disk for retrieval efficiency. Each query inference (retrieval) typically takes 2-5 seconds depending on the strategy.

4.7.2 Hyperparameters and Configuration

- **ColQwen2:** Pretrained weights from Hugging Face (`Qwen/ColQwen2-v1.0`), used as-is without fine-tuning.
- **docTR OCR:** Pretrained model (`doctr-pytorch`) applied to all documents for text extraction.
- **BM25:** Standard hyperparameters ($k_1 = 1.5, b = 0.75$).
- **RRF:** Constant $K = 60$ for reciprocal rank fusion.
- **VLM Re-ranking:** Top-$N = 20$ candidates re-scored by Qwen2-VL (native temperature $= 0$ for deterministic scoring).

4.7.3 Evaluation Metrics

Retrieval performance is evaluated using:

- **Recall@K** ($K \in \{1, 5, 10\}$): Proportion of queries for which the ground-truth document appears in the top-K ranked list.
- **Mean Reciprocal Rank (MRR)**: Average of 1/rank for each query's ground-truth document position (capped at rank 1000).
- **Normalized Discounted Cumulative Gain (NDCG@10)**: Ranking quality metric that weights higher-ranked correct results more heavily than lower-ranked ones.

5 Experiments and Results

5.1 Retrieval Performance Comparison

Table 1 summarizes retrieval performance across all four strategies evaluated on the 2,000 validation queries against the 10,189 deduplicated corpus images. We report Recall@1, Recall@5, Recall@10, Mean Reciprocal Rank (MRR@10), and Normalized Discounted Cumulative Gain (NDCG@10).

Strategy	**Recall@1**	**Recall@5**	**Recall@10**	**MRR@10**	**NDCG@10**
E0: ColQwen2 Dense (MaxSim)	0.3775	0.5630	0.6295	0.4580	0.4992
E1: BM25 over docTR OCR	0.1914	0.2988	0.3471	0.2514	0.2686
E2: RRF (E0 + E1)	0.3585	0.5330	0.5925	0.4189	0.4602
E3: VLM Re-rank (subset 500)	0.2660	0.5060	0.6100	0.3696	0.4269

Bång 1: Retrieval performance across training-free retrieval strategies on 2,000 validation queries. E0 establishes the dense baseline using ColQwen2 MaxSim; E1 introduces lexical matching via BM25 over docTR-extracted OCR text; E2 combines both signals via Reciprocal Rank Fusion (RRF); E3 applies VLM-based semantic re-ranking to E0's top-10 candidates (evaluated on 500 queries due to computational cost).

5.1.1 Dense Retrieval (E0 — ColQwen2 MaxSim)

E0 achieves the strongest absolute performance across all metrics, with Recall@1 = 37.75% and NDCG@10 = 49.92%. This indicates that the late-interaction MaxSim scoring computed over ColQwen2's patch-level embeddings is well-aligned with the multimodal structure of DocVQA documents, capturing both visual layout and textual content without requiring OCR preprocessing.

5.1.2 Sparse Retrieval (E1 — BM25 over docTR OCR)

E1's performance is substantially lower than E0 (Recall@1 = 19.14%, NDCG@10 = 26.86%), indicating that OCR-extracted text alone provides insufficient signal for document retrieval. This gap reflects two factors: (1) docTR OCR errors introduce noise into the lexical index, degrading exact-match queries; and (2) the absence of visual and spatial cues that carry semantic meaning in forms, tables, and layouts limits lexical-only retrieval's effectiveness. However, E1's non-zero performance demonstrates that sparse retrieval retains complementary value — it correctly retrieves documents on queries where lexical overlap is strong (e.g., proper nouns, specific numbers).

5.1.3 Rank Fusion (E2 — Reciprocal Rank Fusion)

E2 fuses the dense (E0) and sparse (E1) ranked lists using RRF with $K = 60$. Despite E1's weakness, fusion achieves measurable results but falls short of E0 alone: Recall@1 = 35.85% (1.9 percentage points vs E0), MRR@10 = 0.4189 (8.53% vs E0), and NDCG@10 = 0.4602 (7.8% vs E0). This partial degradation is noteworthy: although E2 performs better than E1, the sparse retriever's lower quality rankings introduce noise when fused with E0's high-confidence

scores. RRF's rank-based approach mitigates magnitude-scale mismatches, but the fundamental low precision of E1 prevents E2 from surpassing the dense baseline. The result suggests that complementarity between dense and sparse retrieval is limited in this domain when sparse quality is poor.

5.1.4 VLM Re-ranking (E3 — Qwen2-VL on E0 Top-10)

E3 re-ranks E0's top-10 candidates using Qwen2-VL's confidence in document-query relevance. Evaluated on a subset of 500 queries (see Section 5.2), E3 achieves Recall@10 = 61.00%, but lower Recall@1 = 26.60% (8.0% vs E0 subset) and MRR@10 = 0.3696 (5.5% vs E0 subset). The VLM re-ranker frequently demotes the top-1 candidate despite it being correct, suggesting that its semantic judgment differs from ColQwen2's patch-level matching and sometimes contradicts dense retrieval's ranking, introducing noise rather than refinement in the top-1 regime.

5.2 Latency–Accuracy Trade-off

VLM-based re-ranking offers potential accuracy gains at the cost of significant inference latency, motivating analysis of the latency–accuracy frontier.

5.2.1 Inference Latency

Per-query inference times on a single NVIDIA T4 GPU (16 GB) are as follows:

- **E0 (Dense MaxSim):** ≈ 2–3 seconds per query, dominated by corpus scanning across 102 shards; amortized over 2,000 queries ≈ 3.33 ms/query after corpus embeddings are cached.
- **E1 (BM25):** ≈ 5–10 ms per query, negligible; BM25 is a fast, index-based method.
- **E2 (RRF):** ≈ 10–20 ms per query, dominated by merge and re-scoring logic (sum of E0 latency + E1 latency in practice, but E0 dominates).
- **E3 (VLM Re-rank on top-10):** ≈ 200–500 ms per query, or **2–5 seconds per query** for 10 candidates when averaged across batch size 1 and model load time. At this rate, re-ranking the full 2,000 validation queries would require $2,000 \times 5$ s/3600 $=\approx$ 2.8 hours on a single T4, prohibitive for frequent re-runs during development.

  Due to computational constraints, E3 was evaluated on a random subset of 500 queries (seed 42), reducing cost to ≈ 40 minutes. **This design choice is crucial for reproducibility**: E3 metrics are not directly comparable to E0/E1/E2 full 2,000-query results because subset distribution may differ.

5.2.2 Accuracy–Latency Trade-off Analysis

Table 1 shows that the dense ColQwen2 baseline (E0) achieves the strongest retrieval performance overall, obtaining Recall@1 of 37.75%, Recall@10 of 62.95%, MRR@10 of 0.4580, and NDCG@10 of 0.4992.

Although VLM-based re-ranking (E3) introduces an additional semantic reasoning stage, it does not improve retrieval effectiveness in our experiments. On the evaluated subset of 500 validation queries, E3 achieves Recall@1 of 26.60%, Recall@5 of 50.60%, Recall@10 of 61.00%, MRR@10 of 0.3696, and NDCG@10 of 0.4269, all lower than the dense baseline.

These results suggest that ColQwen2's late-interaction MaxSim scoring already provides a strong ranking of candidate documents. Re-ranking with Qwen2-VL occasionally promotes visually or semantically related documents that do not contain the correct answer, reducing top-rank retrieval accuracy. While Recall@10 remains relatively close to the dense baseline (61.00% vs. 62.95%), the substantial drop in Recall@1 and MRR indicates that the VLM frequently reorders correct top-ranked documents to lower positions.

Overall, the additional semantic reasoning capability of the VLM does not translate into improved retrieval quality under the current prompting strategy.

5.2.3 Practical Guidance

Based on the experimental results, different retrieval strategies are suitable under different operational constraints:

- **Dense Retrieval (E0).** ColQwen2 provides the best overall retrieval performance while maintaining reasonable inference latency. It consistently achieves the highest Recall@1, Recall@5, Recall@10, MRR, and NDCG among all evaluated strategies, making it the preferred choice for practical multimodal document retrieval systems.

- **Sparse Retrieval (E1).** BM25 over OCR text is computationally inexpensive and remains useful as a lightweight lexical baseline. However, its retrieval effectiveness is considerably lower because OCR errors and the lack of visual layout information limit its ability to distinguish visually similar documents.

- **Hybrid Retrieval (E2).** Reciprocal Rank Fusion does not outperform the dense baseline in our experiments. The relatively weak BM25 rankings introduce additional noise into the fused ranking, preventing RRF from fully exploiting the complementary strengths of dense and sparse retrieval.

- **VLM Re-ranking (E3).** Although VLM re-ranking provides richer semantic reasoning, its computational cost is substantially higher while yielding lower retrieval accuracy than the dense baseline. Under the current prompting strategy, the additional inference time is not justified by the observed retrieval performance.

6 Conclusion and Future Work

6.1 Conclusion

This project presented a systematic comparison of four training-free retrieval strategies for multimodal document retrieval on the DocVQA benchmark using ColQwen2 as the dense retrieval backbone. The proposed pipeline combines dense retrieval, sparse OCR-based retrieval, rank

fusion, and VLM-based re-ranking within a unified experimental framework, allowing their strengths and limitations to be evaluated under identical conditions.

Experimental results show that ColQwen2 dense retrieval consistently achieves the best overall performance, obtaining the highest Recall@1 (37.75%), Recall@5 (56.30%), Recall@10 (62.95%), MRR@10 (0.4580), and NDCG@10 (0.4992) among all evaluated methods. This demonstrates that late-interaction retrieval over visual patch embeddings is highly effective for multimodal document retrieval without requiring OCR preprocessing.

The OCR+BM25 baseline (E1) provides a useful lexical retrieval signal but is significantly limited by OCR quality and the inability to capture document layout and visual semantics, achieving only 19.14% Recall@1 and 0.2686 NDCG@10. Although Reciprocal Rank Fusion (E2) combines dense and sparse rankings without additional training, it does not outperform the dense baseline (35.85% Recall@1 vs. 37.75%) because the sparse retriever introduces noisy rankings. Similarly, VLM-based re-ranking (E3) incurs substantially higher inference cost (2–5 seconds per query) while failing to improve retrieval accuracy (26.60% Recall@1 on the 500-query subset) under the current prompting strategy.

Overall, the experiments indicate that a strong pretrained dense retriever such as ColQwen2 already provides highly competitive retrieval performance for DocVQA. Additional training-free retrieval components do not necessarily improve accuracy, emphasizing that the effectiveness of hybrid retrieval largely depends on the quality of the complementary retrieval signals. For practical multimodal RAG systems, ColQwen2 dense retrieval (E0) is the recommended choice, providing the best balance of retrieval accuracy and computational efficiency.

6.2 Limitations

Despite the promising results, this project has several limitations.

First, the experiments rely entirely on pretrained models without any task-specific fine-tuning. While this enables a fair comparison of training-free retrieval strategies, it also limits the potential performance improvements that could be achieved through supervised optimization.

Second, the sparse retrieval component depends heavily on OCR quality. Recognition errors produced by docTR directly affect the BM25 index, reducing lexical matching performance, especially for handwritten documents, noisy scans, or complex layouts. During development, we explored alternative OCR models to improve E1's performance: **EasyOCR** produced outputs with unacceptably low accuracy compared to docTR, while **PaddleOCR** encountered numerous library compatibility and version conflicts with PyTorch and CUDA, making reproducible integration infeasible within the project timeline. As a result, experiments proceeded with docTR as the primary OCR backend, despite its acknowledged limitations.

Third, VLM-based re-ranking is computationally expensive. Due to limited GPU resources and long inference time, experiments were conducted on only a subset of the validation queries, making the evaluation less comprehensive than the other retrieval strategies.

Another practical limitation encountered during this project was the software environment. Running ColQwen2, docTR, Qwen2-VL, and their associated dependencies required different versions of PyTorch, CUDA, Transformers, and Flash-Attention. Resolving compatibility conflicts between these libraries required considerable engineering effort and restricted experimentation with alternative model configurations.

Finally, due to hardware and time constraints, this work does not investigate retriever fine-tuning, OCR model optimization, or learned retrieval fusion methods. Therefore, the reported results primarily reflect the capabilities of existing pretrained models under a training-free setting.

6.3 Future Work

Several directions can be explored to further improve multimodal document retrieval.

First, future work could investigate trainable retrieval models. Fine-tuning ColQwen2 on DocVQA or other document retrieval datasets using contrastive learning may improve document representations and reduce retrieval errors on visually similar documents, potentially outperforming the training-free baseline.

Second, retrieval performance may benefit from improving the sparse retrieval component. More robust OCR systems, layout-aware text extraction, or document structure analysis could produce higher-quality lexical indexes, potentially increasing the effectiveness of hybrid retrieval methods and enabling RRF fusion to deliver gains over dense retrieval alone.

Third, VLM-based re-ranking deserves further investigation. Future work should explore alternative VLM architectures, improved prompting strategies to better align with visual patch-level semantics, or uncertainty-driven re-ranking approaches that apply VLMs only to queries where the dense retriever's confidence is low, reducing computational overhead.

Future research may also explore more advanced retrieval fusion techniques beyond Reciprocal Rank Fusion, including score calibration, learning-to-rank approaches, or neural fusion models that dynamically weight dense and sparse retrieval signals based on query characteristics.

Finally, the proposed retrieval system could be extended into a complete multimodal Retrieval-Augmented Generation (RAG) framework. In such a system, retrieved document images would serve as external knowledge for a vision-language model, enabling accurate question answering, document summarization, and information extraction across large-scale document collections.